The Pressure-Supported Excess: What a Dark-Energy-Tied Galaxy Law Cannot Yet Explain, and a Native-Input Shear Rerun

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Abstract

We report one more day of pre-registered tests of a gravity framework whose galactic law is $g_N = g^2/(g + a_0)$ with $a_0 = \frac{1}{2}c\sqrt{G\rho_{\rm DE}}$ (the $\frac{1}{2}$ is fitted). Its extra gravitating mass is a collisionless "cold energy" component, not a particle. Every lane froze its criteria before computing and ran a mutation control. Post-hoc amendments are disclosed, and both a_0 footings (9.36×10^{-11} and $1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$) are reported and never pooled.

1. **Shear.** Rerun from the framework's own early-universe inputs, the canonical-footing structure stays CONSISTENT with raw KiDS-1000 shear against its matched control. The size of its lead is set mainly by the lensing amplitude, and neither arm fits well in absolute terms. The alternative-footing native rerun is still running.
2. **CMB lensing.** On raw Planck PR4 lensing (L 8–400) the finished runs cannot be told from their controls. They are CONSISTENT only under a post-hoc amendment, and the test has little power. A sealed forecast finds that S4-class lensing at $L \approx 1000$–2000 could separate them; ACT-like data and Planck's multipole range cannot.
3. **The pressure-supported excess.** Every system with an excess over the law (dwarfs, spheroidal lenses, cluster cores) is pressure-supported, and no rotating system has one. But no single kinematic measure or local density sets the amounts, none of five candidate selectors fits all the facts, and the record's own cold-energy equations cannot select on kinematics at all. The cluster-core excess is not a baryon-census error. A stellar-IMF correction removes the red-lens excess if discs are not bottom-heavy; after it, one velocity-scale selector fits the galaxy-scale facts, conditionally, but not the clusters. The dwarf excess survives every reframing tried. A fossil a_0 helps only under the rival $a_0 \propto H(z)$.

1 The native-input shear rerun

In the previous note the framework's combined particle-mesh engine ("theory-v2": per-host supply, condensed source, gas expulsion; 1024^3 particles, $100\,\mathrm{Mpc}/h$) was CONSISTENT with raw KiDS-1000 cosmic shear on both footings, conditional on inherited ΛCDM-shaped initial conditions and background. A pre-registered addendum (CFG654 A2) reran the canonical footing as run NC1x from the framework's own early-universe inputs: the Planck TT,TE,EE+lowE values of ω_b, ω_c, A_s, n_s and θ_*, passed through CAMB in the era before the law switches on, with the framework's own turnaround and catchment. Its control, NCS0, is the same pipeline on the same initial conditions with plain Poisson gravity. The frozen scorer was not changed, and its mutation control ran first and bites.

survey	points	$\chi^2_{\rm min}$ control	$\chi^2_{\rm min}$ framework	$\Delta\chi^2$	robustness range	class
KiDS-1000	108	154.76	144.68	−10.08	−6.13 to −13.78	CONSISTENT
DES Y3	78	93.99	91.78	−2.21	−0.79 to −4.97	NOT DIAGNOSTIC

The pre-declared reading is HOLDS: NC1x is CONSISTENT and $\Delta\chi^2 \leq 0$ against its control. Three caveats limit what this means.

- **The size is amplitude-driven.** A re-scoring lane (CFG909) varied the matter density used in the shear projection over $\Omega_m = 0.2806$–0.3308. KiDS stays CONSISTENT in every variant (class ROBUST), but $\Delta\chi^2$ runs from −3.11 to −15.80, and with the framework's own covariance it is −2.94. The framework's run has less small-scale power than its control (ratio 0.972, 0.961 and 0.949 at $k = 0.3$, 0.5 and $1\,h/\mathrm{Mpc}$, $z = 0$). On the inherited-input runs CFG909 traced the lead mainly to such a lower lensing amplitude (a 3–6% power deficit), not to a shape preference. We therefore quote the result as CONSISTENT, not as a $\Delta\chi^2 = -10$ preference.
- **Neither arm fits well in absolute terms.** The framework reaches $\chi^2_{\rm min} = 144.68$ on 108 KiDS points ($p = 0.024$ on the data part of χ^2), and its control 154.76 ($p = 0.008$). Both are worse than the inherited-input run (136.6). The framework beats its control by suppressing small-scale power.

- **Inputs still shared with standard cosmology:** the Planck parameter column (from a base-ΛCDM chain), H_0 from θ_*, massless neutrinos, the empirical retained-fraction census, the cooling inputs, and the survey nuisances and covariances.

So the native rerun lifts the inherited-input condition on the canonical footing only. The alternative-footing native rerun (NC2x; addendum frozen before the job started) is running; no number from it is reported here, and until it finishes a two-footing native pass is not established. Its pre-declared reading will be reported whichever it is.

2 CMB lensing

CFG917 asked whether the framework's gravitating field shows up in CMB lensing. Its criteria were frozen after NC2x started and before any NC2x output. It has two parts.

- **Relative check.** An earlier checklist statistic (ΔA on Planck 2018 band weights, L 8–400, with $2\sigma = 0.056$) gives -0.0018, -0.0017 and -0.0016 for NC1x, T1x and T2x. All three are SAFE.
- **Raw bandpowers.** Each run's own Limber $C_L^{\phi\phi}$, built from its measured power spectrum, was scored against the raw Planck PR4 lensing bandpowers (CMB-marginalised, L 8–400, 9 bands, published covariance and linear correction).

pair	footing / inputs	χ^2 framework	χ^2 control	$\Delta\chi^2$	as frozen	under A1
NC1x vs NCS0	canonical / native	9.265	9.541	-0.276	NOT DIAGNOSTIC	CONSISTENT
T1x vs P0	canonical / inherited	9.555	9.643	-0.088	NOT DIAGNOSTIC	CONSISTENT
T2x vs P0	alt. / inherited	9.558	9.643	-0.085	NOT DIAGNOSTIC	CONSISTENT

- **The amendment.** As frozen, every pair is NOT DIAGNOSTIC, because one gating check failed. A post-hoc amendment (A1), written before any T1x, T2x or NC2x bandpower number, found that the check tested a wrong premise about the CMB-marginalised likelihood's fiducial correction. It replaced the check with a data-only one that the code passes, reproducing the likelihood package's unit-test values. Under A1 all three pairs are CONSISTENT and the gate passes; that pass rests on A1. Every robustness member stays below 1.6 in $|\Delta\chi^2|$.
- **Low power.** About 70% of the 8–400 signal comes from scales larger than the box, where both arms share the same linear spectrum by construction. The framework moves the bandpowers by at most 1%, against a PR4 amplitude error of about 2.4%. A candidate already excluded by KiDS (CFG596, KiDS $\Delta\chi^2 = +68.7$) reaches only +1.88 here. So CMB lensing at $L \leq 400$ does not contradict the shear pass, and it is not strong evidence for it. NC2x will be scored blind.

A sealed forecast. CFG927 sealed a Fisher forecast of each run against its control. It uses a Gaussian covariance and quadratic-estimator reconstruction noise for three survey models; every run supports the computation to $L = 3000$. The table and its hash are committed. A forecast is not a measurement.

Fisher $\Delta\chi^2$ (range over T1x, T2x, NC1x)	$L \leq 400$	$L \leq 1000$	$L \leq 2000$	$L \leq 3000$
ACT-DR6-like	0.06–0.08	0.73–0.91	1.73–2.11	1.83–2.22
SO-like	0.26–0.33	4.25–5.25	10.23–12.40	11.84–14.30
S4-like	0.63–0.77	29.42–35.44	133–157	186–219
noiseless (context)	0.82–1.00	81.45–95.59	1639–1819	7428–8044

- **Who can see it.** With $\Delta\chi^2 \geq 9$ as DISCRIMINATING, the ACT-like survey stays blind to $L = 3000$. The SO-like survey needs multipoles to $L \approx 2000$ (10–12), and that class depends on the robustness rule. The S4-like survey discriminates from $L \approx 700$ and robustly by $L \leq 1000$ (29–35). Even a noiseless measurement at $L \leq 400$ gives only 0.82–1.00, so CFG917's low power is the multipole range, not Planck's noise. The KiDS-excluded candidate, by contrast, is discriminating in every survey model (21.37 for the ACT-like survey at $L \leq 763$).
- **Caveats.** The continuation above $z = 1$ carries 45–78% of the theory-v2 difference; the noise models are optimistic (forecast amplitude errors are 0.68 and 0.55 of the published Planck and ACT values) and mostly illustrative; NC2x is not forecast.

High-L lensing at S4-class depth is therefore the first CMB channel that could separate theory-v2 from its control.

3 The pressure-supported excess

The previous note left two open excesses over the law at low acceleration: spheroidal (red) isolated lenses, and Milky Way and M31 satellite dwarfs. Cluster cores carry a third (about $\times 1.8$ in the core, from earlier lanes). The lanes below treat the three together.

3.1 The pattern

CFG919 placed every class at matched $g_{\rm bar}$ against one kinematic measure, the random fraction of the baryons' kinetic energy, $f_P = 3\sigma^2/(v^2 + 3\sigma^2)$. Class means are over 0.5-dex cells in $g_{\rm bar}$; the excess is $\log_{10}(g_{\rm obs}/g_{\rm law})$.

class	f_P	mean excess, dex (canonical)	mean excess, dex (alternative)
83 Local Volume dwarfs	1.0	$+0.372 \pm 0.044$	$+0.332 \pm 0.044$
eRASS1 clusters (shear stacks)	1.0	$+0.243 \pm 0.033$	$+0.207 \pm 0.033$
isolated lenses, Sérsic $n > 2$	0.8 (proxy)	$+0.127 \pm 0.029$	$+0.088 \pm 0.029$
isolated lenses, Sérsic $n < 2$	0.2 (proxy)	$+0.003 \pm 0.040$	-0.036 ± 0.040
SPARC small discs	0.075	$+0.059 \pm 0.037$	$+0.025 \pm 0.036$
SPARC large discs	0.012	$+0.001 \pm 0.022$	-0.023 ± 0.022

- **The ordering is real.** At matched $g_{\rm bar}$, pressure-supported classes sit above rotation-supported ones. The ordering statistic is $T = +0.528$ with permutation $p = 0.002$. Spheroidal lenses sit above disc-like lenses by $\Delta E = +0.29$ in natural-log units ($z = +3.5$). No rotating cell carries an excess.
- **There is no single kinematic form.** A linear form $E = Af_P$ gives $\chi^2 = 89.4$ for 32 degrees of freedom on the canonical footing and 93.5 on the alternative ($p < 10^{-6}$ on both), and a step at $v/\sigma = \sqrt{3}$ does no better. The dwarfs carry 3–5 times the spheroidal lenses' excess at the same $g_{\rm bar}$, and at $f_P \approx 1$ the clusters' excess rises with $g_{\rm bar}$ while the dwarfs' falls.
- **Class and fragility.** The class is SELECTOR CONTRADICTED on both footings, as frozen and under a post-hoc amendment (identical results). It is FRAGILE: doubling the per-cell systematic floor to 0.10 dex turns the canonical footing SUPPORTED and the alternative NOT DIAGNOSTIC. Lens kinematics are a Sérsic-index proxy, the dwarf f_P is a class value, early types with measured dynamics are not tested, and globular clusters carry no class weight.
- **Density does not order it either (CFG914).** At the same local density, dwarfs carry an excess where SPARC discs do not, and cluster cores carry one where blue lenses do not. Clusters rise and red lenses fall with density. Local density is excluded as the unifying cause on both footings and in all ten variants.

3.2 The record's equations cannot select on kinematics

In the record's cold-energy system (CFG638) the baryons enter only through their density: in the potential, in the settling clock, in the law's phantom target and in the region indicators. No baryon velocity or stress appears. In the stationary state the cold energy settles onto the phantom target, which in spherical symmetry is fixed by the enclosed baryonic mass. A rotating system and a dispersion-supported one with the same enclosed baryonic mass therefore settle to the same $g(r) = g_{\rm law}$: the equations cannot order an excess by kinematic state, and they produce no excess over the law in equilibrium. CFG888 reached the same point from the action: the law's source is the baryons in every formulation in the record, and a cold-energy clump has no self-gravity in the canonical action. Any kinematic selector is therefore a new posit: a coupling to the baryons' stress (with a new velocity scale), a retention rule that depends on kinematic state, or a non-spherical geometric effect not yet calculated.

3.3 The dwarf excess is not environmental

The excess is not tied to gas lost by stripping (CFG860, NOT GAS on both footings): isolated gas-free and gas-bearing dwarfs carry it too. With gas counted, all 11 isolated Local Group dwarfs sit $+0.194 \pm 0.077$ dex over the law and -0.100 ± 0.091 dex below the satellites (CFG874), UNDECIDED between a satellite property and a small-system property; the 11 are the whole supply. There is no significant trend with host distance (CFG875, slope -0.190 ± 0.150 dex per dex, UNDECIDED).

3.4 Cluster cores are not a census error

CFG922 re-checked the cluster-core excess ($A_{\rm MID} = 1.91$ canonical, 1.75 alternative, for 318 eRASS1 clusters) against every baryon component on disk.

- **What is needed.** To bring $A_{\rm MID}$ to 1 the law needs extra concentrated baryons inside 0.5 Mpc equal to 2.24 (canonical) or 1.84 (alternative) times the hot gas there.
- **What cannot supply it.** Stars would need 32.1 / 26.6 times the top of the measured stellar bracket. A hot-gas renormalisation would need 3.68 / 3.21, against an upper end of 1.50 measured from 68 cross-matched clusters. Radius convention, hydrostatic bias, clumping and Eddington bias cannot (three have the wrong sign), nor can cold atomic gas or a bottom-heavy IMF.
- **All at once.** With every testable component at its upper end the core still sits at 1.49 / 1.37 ($Z = +5.2$ / $+4.3$). Any baryon fix that removes the core excess also breaks the outer agreement.
- **Untestable.** Intracluster light and BCG molecular gas have no tables on disk, and either would need 22–27 times the stellar bracket.
- **Class.** EXCESS STANDS on both footings and both kernels, as frozen and under a post-hoc census amendment that moved no class. The core excess is a property of the extra mass, not of the census.

3.5 Five candidate selectors all fail

CFG921 graded five candidate new selectors on all the facts above, with a trials factor of 5. Four have one constant each, fixed once on the 83 dwarfs; the fifth is a geometric effect with no constant. P1, a baryon-stress coupling, cannot even be calibrated: there is no equilibrium at the dwarf amplitude. P2, velocity-space capture, fails on clusters ($z = -7.3$ / -6.2) and red lenses (-4.4 / -3.0). P3, late binding plus escape, fails on clusters, red lenses and the dwarf size split. P4, phase-space-density capture, fails on clusters (-7.3 / -6.2); its canonical red-lens miss (3.8σ at the least severe corner) does not reach the trials-factor bar. P5, the QUMOND geometry of spheres versus discs, has the right sign for spheres but is about 40 times too small. None survives on either footing. The reason is structural. Dwarfs ($\sigma \sim 10$ km/s) and red lenses ($\sigma \sim 100$ km/s) need a 3–5× suppression between them. Selectors with a velocity scale give $\gtrsim 100\times$, and scale-free ones give $1.25\times$. The cluster excess rises with $g_{\rm bar}$, where every capture or latch posit gives zero.

3.6 The dwarf route and its conditions

The dwarf excess has been modelled as a retained cold-energy clump. Three lanes limit that picture, and mobility rules do not rescue it.

- **No universal clump velocity (CFG880).** Across 83 dwarfs the intrinsic scatter in the log of the clump velocity is 0.380 (95%: 0.281–0.510, canonical; the alternative also gives 0.380), about $\times 1.46$ per dwarf. NOT UNIVERSAL on both footings and in all six variants.
- **The clump relations are not derived (CFG888).** The clump scale relations used in earlier fits do not follow from the action. A finite Newtonian clump needs a truncation that adds a parameter.
- **The gate does not keep the clump (CFG889).** With orbits mixing, the clump erodes in all eight cells, with a gated half-life of 105–148 Myr/α. This lane is inconclusive as frozen, because one check failed.
- **One local law (CFG913).** No single local mobility law, in any of 42 pre-declared forms, reconciles mergers, galaxies and retained clumps. A galaxy's dense inner region must stay mobile to finish settling.
- **A history label (CFG915).** Making cold energy bound at formation immobile, and later arrivals mobile, only moves the obstruction to cluster outer profiles. It is CONTRADICTED under both label rules.

Table 1: Six small reframings of the pressure-supported excess.

reframing	class	effect (both footings share each class)
R1 law-based Jeans dispersion, not the Wolf estimator	NO EFFECT	dwarf excess $+0.372 \rightarrow +0.369$ dex (alt. $+0.332 \rightarrow +0.329$); strongly radial orbits give at most -0.114 dex
R2 cold energy in the law's own argument	WORSENS	literal form ill-posed; the primary form raises every class by 0.1–1.8 dex; both well-posed forms break rotators
R3 the framework's external-field effect	WORSENS	satellites' excess up by $+0.48$ dex on average; dwarf class $+0.372 \rightarrow +0.654$ dex
R4 fossil a_0 from the formation epoch	see text	framework $a_0(z)$: none or wrong sign; rival $a_0 \propto H(z)$: PARTIAL
R5 IMF α–σ relation for old spheroids	RESOLVES	red-lens excess $+0.127 \rightarrow +0.048$ dex (alt. $+0.088 \rightarrow +0.009$); split z 2.53 $\rightarrow$ 0.92
R6 lensing estimator, round versus flat lenses	NO EFFECT	shift -0.004 (ln); the full estimator gives -0.023, in the wrong direction

- **Release (CFG916).** Adding release beyond the tidal radius is CONTRADICTED as posed. It is UNDERDETERMINED only with a new capped-capture rule.

What survives is one dwarf route. It needs cold energy bound at formation to be fully immobile, dwarfs to have lost a median $\times 4.32$ of their baryons after formation, and binding to come before that loss. Three lanes test its conditions.

- **Chemistry (CFG920): UNDERDETERMINED on both footings.** The sample's median required loss is 3.81 (canonical) or 3.18 (alternative) for the 20 dwarfs with metallicities. The chemistry holds this at a low iron yield and excludes it at a high one. Per dwarf (canonical, 1σ): 1 compatible, 15 underdetermined, 4 contradicted. The metal-poor gas-free dSphs strain it unless the iron yield is low, winds are metal-enhanced, or binding is late.
- **Clock (CFG923): UNDERDETERMINED overall, because the footings disagree.** The framework's own formation clock binds too early for most dwarfs that need late binding: the universal clock is CONTRADICTED on the canonical footing and UNDERDETERMINED on the alternative, and the orbital clock is CONTRADICTED on both. Late binding survives only as a new posit, a trigger 4.9–12.7 Gyr after star formation begins in some dwarfs and not others. The hardest cases are isolated field dwarfs, so the trigger would have to be internal.
- **Receiver (CFG918).** For Draco, Sextans, Fornax and Leo I, the route needs somewhere to put the energy and angular momentum it sheds. A bound outer envelope is excluded for Sextans in one mass branch, and for Draco and Sextans when transfer starts at the catchment radius. With escape allowed, the capacity is there except in one indeterminate cell. Only an escaping receiver remains; this tests necessary feasibility, not formation.

3.7 Small reframings

Two lanes tried six reframings that add no new constant (CFG925: R1–R3; CFG926: R4–R6; Table 1).

- **The dwarf excess is not an estimator artefact** (R1). Self-sourcing breaks rotators (R2).
- **An external-field effect only deepens the satellite excess** (R3). The record's dwarf numbers are field-free, so the quoted excess is a lower bound if the framework has such an effect.
- **The red-lens excess may be a stellar-IMF effect** (R5), on all nine corners and both footings. This is conditional: it holds only if discs stay near a Chabrier IMF (with the same relation applied to discs it is PARTIAL), and on the earlier colour-window statistic the canonical removal is 49%, or 45% under the record's own IMF ceiling (post hoc), both of which would read PARTIAL. It has no effect on clusters and leaves the dwarfs unchanged.
- **The fossil** a_0 (R4) supposes that pressure-supported systems keep the a_0 of their formation epoch. The framework's flat a_0 gives exactly zero. Its dark-energy-tracking $a_0(z)$, with the DESI fits, gives the wrong sign: dwarfs would sit about 0.07 dex below the law. Only the rival $a_0 \propto H(z)$ gives a positive fossil. It matches the dwarf mean ($+0.81$ against $+0.86$ in natural-log units, canonical), but overshoots the red lenses and gives 28% of the cluster excess with the wrong slope. We report this as evidence about $a_0(z)$, not

as a rescue: if the excess were a fossil, it would point to the rival and away from the framework's flat or dark-energy-tracking a_0. The rival's own failures in the record stand.

A conditional re-grade (CFG928). With the red-lens cells IMF-corrected as in R5 and the clusters treated as a separate class, CFG921's five selectors were re-graded with no constant refitted (trials factor 20). P2, velocity-space capture ($v_2 = 6.95$ / 6.46 km/s), then SURVIVES on the galaxy-scale facts on both footings, but still FAILS once the clusters are included ($z = -7.3$ / -6.2): a separate cluster explanation would have to supply $\times 1.75$ / $\times 1.61$, rising with $g_{\rm bar}$. P4 is UNDERDETERMINED; P1, P3 and P5 fail. The survival is weak. It is post-hoc-motivated and conditional on discs staying near Chabrier. It is FRAGILE (without the earlier $+0.123$ dex stellar-mass correction it fails on the canonical footing) and drops to UNDERDETERMINED on the canonical footing under weaker IMF variants. It is the best of 20 posit-framings tried, so it is not evidence for P2. A prediction for pressure-supported ultra-diffuse galaxies, which P2 places nearly on the isolated law, is sealed. **In progress:** CFG929, four small reframings of the cluster-core excess (criteria frozen, no ledger results yet).

4 What this establishes, and what it does not

- **What holds, conditionally.** The canonical shear pass survives the framework's own early-universe inputs (CONSISTENT; amplitude-driven; neither arm a good absolute fit). CMB lensing at $L \leq 400$ does not contradict it (post-hoc amendment; little power). CONSISTENT is a class, not a preference.
- **What is not established.** A two-footing native pass (NC2x is running), and any CMB-lensing evidence for the framework; S4-class high-L lensing is a sealed future test.
- **What is open: the pressure-supported excess.**
  - The empirical pattern is firm: pressure-supported systems carry an excess and rotating ones do not. The record's equations produce no excess in equilibrium.
  - None of these sets the amounts: local density, one kinematic measure, five candidate selectors on all the facts, environment, the dispersion estimator, self-sourcing, an external field, the lensing estimator, or a fossil of the framework's own $a_0(z)$.
  - The cluster-core excess is not a census error. The red-lens excess may be a stellar-IMF effect if discs are not bottom-heavy; after that correction one selector fits the galaxy-scale facts (conditional, post hoc), not the clusters.
  - Nothing tested explains the dwarf excess. The one dwarf route left needs an immobile bound component, a median $\times 4.3$ baryon loss, a late internal binding trigger and an escaping receiver. Each is a new posit or a strained condition.
- **What is not derived.** $\kappa = \frac{1}{2}$, the cold-to-baryon supply ratio and the retained fraction remain inputs.

Disclosures. Post-hoc amendments: CFG917 A1 replaced a gating check that had a wrong premise, after the NC1x bandpower run and before any T1x, T2x or NC2x bandpower number, and the gate pass rests on it. CFG919 A1 came after the frozen injection control failed and before the main run; CFG914 A1 revised an injection control after the frozen one failed; neither changed a class. CFG921 A1 (numerical fix, root choice, added diagnostic) and CFG925 A1 (solver tolerance; no value moved) came after check-stage runs, before the mutation and main runs. CFG926 A1 clarified controls and corners before any run (no class rule changed). CFG922 A1 amended the census and replaced an under-powered control (no class moved). CFG918 A1 is reported only. Not blind: CFG919, CFG920, CFG923 and CFG926 saw record numbers before freezing, and say so. Controls: CFG913's frozen mutation control does not bite (the lane is over-determined); CFG889 is inconclusive as frozen.

Public repository lanes (frozen criteria, scripts, outputs) under `campaign_fresh_gravity/`: CFG654 (addenda A2, A3), CFG909, CFG913–CFG923, CFG925–CFG928, CFG860, CFG873–CFG875, CFG880, CFG888, CFG889. Data: M. Asgari et al., A&A 645, A104 (2021) (KiDS-1000); A. Amon et al., PRD 105, 023514 (2022) and L. F. Secco et al., PRD 105, 023515 (2022) (DES Y3); J. Carron, M. Mirmelstein, A. Lewis, JCAP 09 (2022) 039 (Planck PR4 lensing); Planck Collaboration, A&A 641, A6 and A8 (2020); M. M. Brouwer et al., A&A 650, A113 (2021); F. Lelli, S. McGaugh, J. Schombert, AJ 152, 157 (2016) (SPARC); D. R. Weisz et al., ApJ 789, 147 (2014); the Local Volume Database; E. Bulbul et al., A&A 685, A106 (2024) (eRASS1); K. W. Cavagnolo et al., ApJS 182, 12 (2009) (ACCEPT); S. Ettori et al., A&A 621, A39 (2019) (X-COP); M. Cappellari et al., MNRAS 432, 1709 and 1862 (2013) (ATLAS$^{\rm 3D}$); M. W. Auger et al., ApJ 705, 1099 (2009) (SLACS). J. Wolf et al., MNRAS 406, 1220 (2010); G. Chabrier, PASP 115, 763 (2003). AI assistance: Anthropic's Claude (analysis lanes, review, drafting).